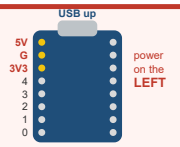
Top view, enlarged for reading

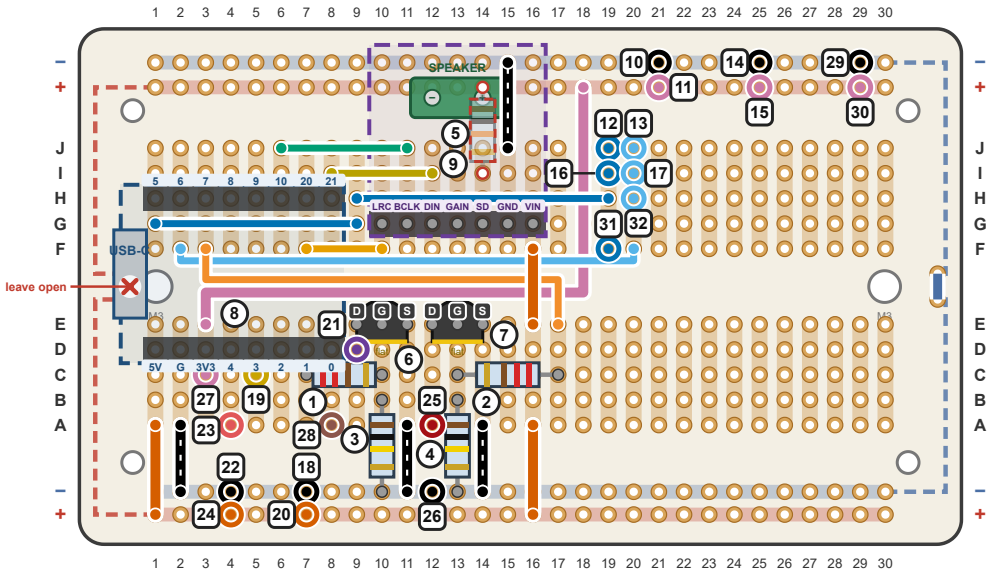
Intake panel, C3 POWER-LEFT — protoboard

TOP VIEW (component side), enlarged, not to scale

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C7, C10
2. RG2 220 Ω: C17, C13
3. RP1 100 kΩ: B10, bottom − rail at number 10
4. RP2 100 kΩ: B13, bottom − rail at number 13
5. RSD1 10 kΩ: top + rail at number 14, I14; fit only if the amplifier's SD pin reads near 0 V in the SD pull-up check
6. Q1 BS170: E9, E10, E11
7. Q2 BS170: E12, E13, E14
8. U1 ESP32-C3 Super Mini: D1-D8, H1-H8
9. U2 MAX98357 amplifier: G10-G16

Device wires

OLED

10. GND: top − rail at number 21
11. 3V3: top + rail at number 21
12. SDA: J19
13. SCL: J20

AIR

14. GND: top − rail at number 25
15. 3V3: top + rail at number 25
16. SDA: I19
17. SCL: I20

YELLOW

18. GND: bottom − rail at number 7
19. HOLD: C5
20. 5V: bottom + rail at number 7
21. LED− Y: D9

RED

22. GND: bottom − rail at number 4
23. DISC: A4
24. 5V: bottom + rail at number 4
25. LED− R: A12

BUZZER

26. GND: bottom − rail at number 12
27. 3V3: C3
28. BUZ: A8

NFC

29. GND: top − rail at number 29
30. 3V3: top + rail at number 29
31. SDA: F19
32. SCL: H20

Nets (wire colours)

- **GND** ground
- **SCL** I2C clock, GPIO 6
- **5V** USB 5 V (power)
- **BCLK** amplifier bit clock, GPIO 10
- **DIN** amplifier data, GPIO 21
- **HOLD** yellow button, GPIO 3
- **LAMP Y** yellow lamp drive, GPIO 1
- **GATE Y** Q1 gate
- **LED− Y** yellow lamp − / Q1 drain
- **SDA** I2C data, GPIO 5
- **BUZ** buzzer drive, GPIO 0
- **3V3** 3.3 V from the C3 (power)
- **LRC** amplifier word select, GPIO 20
- **SD** amplifier SD (on/off)
- **DISC** red button, GPIO 4
- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves
- leave open: fitting a link here would join two rail halves that carry different nets (see the build steps)
- DNP: a part fitted only when needed (see the build list); keep its holes free
- a wire laid along the centre channel (no copper there), under U1 where drawn
- This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

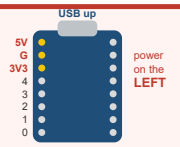
Solder side, enlarged for reading

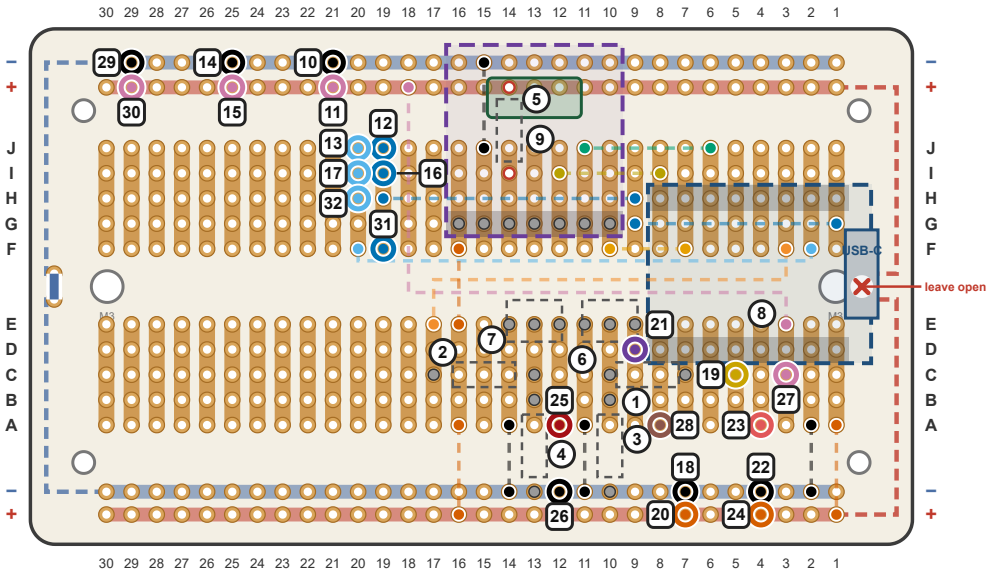
Intake panel, C3 POWER-LEFT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C7, C10
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6. Q1 BS170: E9, E10, E11
7. Q2 BS170: E12, E13, E14
8. U1 ESP32-C3 Super Mini: D1-D8, H1-H8
9. U2 MAX98357 amplifier: G10-G16

Device wires

OLED

10. GND: top − rail at number 21
11. 3V3: top + rail at number 21
12. SDA: J19
13. SCL: J20

AIR

14. GND: top − rail at number 25
15. 3V3: top + rail at number 25
16. SDA: I19
17. SCL: I20

YELLOW

18. GND: bottom − rail at number 7
19. HOLD: C5
20. 5V: bottom + rail at number 7
21. LED− Y: D9

RED

22. GND: bottom − rail at number 4
23. DISC: A4
24. 5V: bottom + rail at number 4
25. LED− R: A12

BUZZER

26. GND: bottom − rail at number 12
27. 3V3: C3
28. BUZ: A8

NFC

29. GND: top − rail at number 29
30. 3V3: top + rail at number 29
31. SDA: F19
32. SCL: H20

Nets (wire colours)

- **GND** ground
- **SCL** I2C clock, GPIO 6
- **5V** USB 5 V (power)
- **BCLK** amplifier bit clock, GPIO 10
- **DIN** amplifier data, GPIO 21
- **HOLD** yellow button, GPIO 3
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- **LED− Y** yellow lamp − / Q1 drain
- **SDA** I2C data, GPIO 5
- **BUZ** buzzer drive, GPIO 0
- **3V3** 3.3 V from the C3 (power)
- **LRC** amplifier word select, GPIO 20
- **SD** amplifier SD (on/off)
- **DISC** red button, GPIO 4
- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

- wire end: solder here
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- (4) a part on the board (see Parts)
- [9] a wire of an off-board device (see Device wires)
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- leave open: fitting a link here would join two rail halves that carry different nets (see the build steps)
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- a wire laid along the centre channel (no copper there), under U1 where drawn

This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

Real size (print at 100 %, "actual size"): lay the board on the drawing

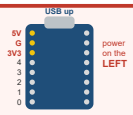
Check the 20 mm bar with a ruler before you rely on the drawing.

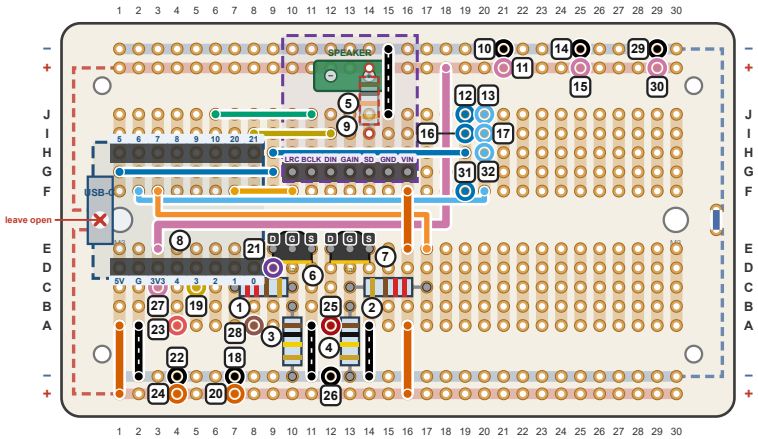
Intake panel, C3 POWER-LEFT — protoboard

TOP VIEW (component side), real size

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C7, C10
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3. RP1 100 kΩ: B10, bottom − rail at number 10
4. RP2 100 kΩ: B13, bottom − rail at number 13
5. RSD1 10 kΩ: top + rail at number 14, I14; fit only if the amplifier's SD pin reads near 0 V in the SD pull-up check
6. Q1 BS170: E9, E10, E11
7. Q2 BS170: E12, E13, E14
8. U1 ESP32-C3 Super Mini: D1-D8, H1-H8
9. U2 MAX98357 amplifier: G10-G16

Device wires

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11. 3V3: top + rail at number 21
12. SDA: J19
13. SCL: J20

AIR

14. GND: top − rail at number 25
15. 3V3: top + rail at number 25
16. SDA: I19
17. SCL: I20

YELLOW

18. GND: bottom − rail at number 7
19. HOLD: C5
20. 5V: bottom + rail at number 7
21. LED− Y: D9

RED

22. GND: bottom − rail at number 4
23. DISC: A4
24. 5V: bottom + rail at number 4
25. LED− R: A12

BUZZER

26. GND: bottom − rail at number 12
27. 3V3: C3
28. BUZ: A8

NFC

29. GND: top − rail at number 29
30. 3V3: top + rail at number 29
31. SDA: F19
32. SCL: H20

Nets (wire colours)

- **GND** ground
- **SCL** I2C clock, GPIO 6
- **5V** USB 5 V (power)
- **BCLK** amplifier bit clock, GPIO 10
- **DIN** amplifier data, GPIO 21
- **HOLD** yellow button, GPIO 3
- **LAMP Y** yellow lamp drive, GPIO 1
- **GATE Y** Q1 gate
- **LED− Y** yellow lamp − / Q1 drain
- **SDA** I2C data, GPIO 5
- **BUZ** buzzer drive, GPIO 0
- **3V3** 3.3 V from the C3 (power)
- **LRC** amplifier word select, GPIO 20
- **SD** amplifier SD (on/off)
- **DISC** red button, GPIO 4
- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves
- leave open: fitting a link here would join two rail halves that carry different nets (see the build steps)
- DNP: a part fitted only when needed (see the build list); keep its holes free
- a wire laid along the centre channel (no copper there), under U1 where drawn
- 20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

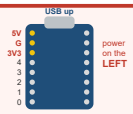
Real size, solder side

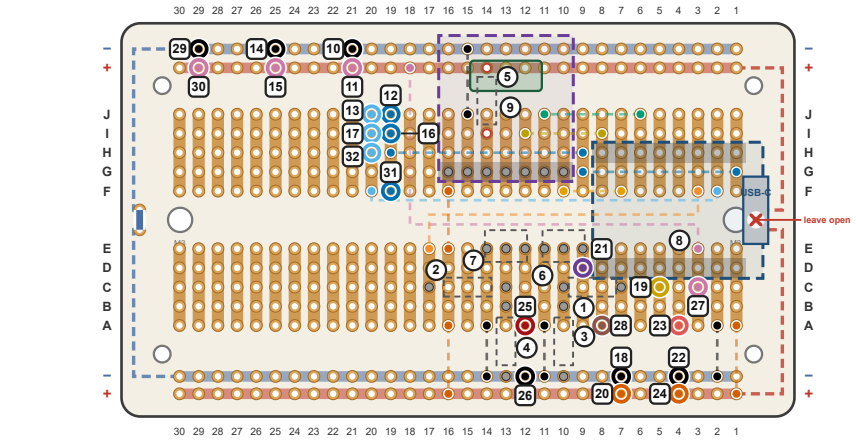
Intake panel, C3 POWER-LEFT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C7, C10
2. RG2 220 Ω: C17, C13
3. RP1 100 kΩ: B10, bottom − rail at number 10
4. RP2 100 kΩ: B13, bottom − rail at number 13
5. RSD1 10 kΩ: top + rail at number 14, I14; fit only if the amplifier's SD pin reads near 0 V in the SD pull-up check
6. Q1 BS170: E9, E10, E11
7. Q2 BS170: E12, E13, E14
8. U1 ESP32-C3 Super Mini: D1-D8, H1-H8
9. U2 MAX98357 amplifier: G10-G16

Device wires

OLED

10. GND: top − rail at number 21
11. 3V3: top + rail at number 21
12. SDA: J19
13. SCL: J20

AIR

14. GND: top − rail at number 25
15. 3V3: top + rail at number 25
16. SDA: I19
17. SCL: I20

YELLOW

18. GND: bottom − rail at number 7
19. HOLD: C5
20. 5V: bottom + rail at number 7
21. LED− Y: D9

RED

22. GND: bottom − rail at number 4
23. DISC: A4
24. 5V: bottom + rail at number 4
25. LED− R: A12

BUZZER

26. GND: bottom − rail at number 12
27. 3V3: C3
28. BUZ: A8

NFC

29. GND: top − rail at number 29
30. 3V3: top + rail at number 29
31. SDA: F19
32. SCL: H20

Nets (wire colours)

- **GND** ground
- **SCL** I2C clock, GPIO 6
- **5V** USB 5 V (power)
- **BCLK** amplifier bit clock, GPIO 10
- **DIN** amplifier data, GPIO 21
- **HOLD** yellow button, GPIO 3
- **LAMP Y** yellow lamp drive, GPIO 1
- **GATE Y** Q1 gate
- **LED− Y** yellow lamp − / Q1 drain
- **SDA** I2C data, GPIO 5
- **BUZ** buzzer drive, GPIO 0
- **3V3** 3.3 V from the C3 (power)
- **LRC** amplifier word select, GPIO 20
- **SD** amplifier SD (on/off)
- **DISC** red button, GPIO 4
- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves
- leave open: fitting a link here would join two rail halves that carry different nets (see the build steps)
- DNP: a part fitted only when needed (see the build list); keep its holes free
- a wire laid along the centre channel (no copper there), under U1 where drawn

20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

Build order and hole table

For a C3 POWER-LEFT module only (USB up: 5V, G, 3V3 at the top of the LEFT row). A power-right module in this layout puts 5 V on a GPIO pin and can destroy it.

Build order (lowest parts first). The numbers are the callouts on the drawing; wire links carry no number:

- Rail link (−): wire from the right link pad, upper to the right link pad, lower. It joins the top and bottom − rails.
- Wire 3V3: E3 to top + rail at number 18 (two bends, along the centre channel, 62 mm).
- Wire SCL: F2 to F20 (two bends, along the centre channel, 48 mm).
- Wire LAMP R: F3 to E17 (two bends, along the centre channel, 43 mm).
- Wire SDA: H9 to H19 (straight, 25 mm).
- Wire SDA: G1 to G9 (straight, 20 mm).
- Wire BCLK: J6 to J11 (straight, 13 mm).
- Wire DIN: I8 to I12 (straight, 10 mm).
- Wire GND: J15 to top − rail at number 15 (straight, 9 mm).
- Wire LRC: F7 to F10 (straight, 8 mm).
- Wire 5V: A1 to bottom + rail at number 1 (straight, 9 mm).
- Wire 5V: A16 to bottom + rail at number 16 (straight, 9 mm).
- Wire 5V: E16 to F16 (straight, 8 mm).
- Wire GND: A2 to bottom − rail at number 2 (straight, 7 mm).
- Wire GND: A11 to bottom − rail at number 11 (straight, 7 mm).
- Wire GND: A14 to bottom − rail at number 14 (straight, 7 mm).
- Part 1, RG1 (220 Ω): C7 to C10, lying flat.
- Part 2, RG2 (220 Ω): C17 to C13, lying flat.
- Part 3, RP1 (100 kΩ): B10 to bottom − rail at number 10, lying flat.
- Part 4, RP2 (100 kΩ): B13 to bottom − rail at number 13, lying flat.
- Part 6, Q1 (BS170): D in E9, G in E10, S in E11, flat face toward the bottom edge (the A side).
- Part 7, Q2 (BS170): D in E12, G in E13, S in E14, flat face toward the bottom edge (the A side).
- Part 8, U1 sockets: two 1x8 female headers in D1-D8, H1-H8. Plug U1 in while soldering to keep them square, then take it out.
- Part 9, U2 socket: a 1x7 female header in G10-G16. Plug U2 in while soldering to keep it square, then take it out.
- Device wires 10-13, OLED: 10 GND in top − rail at number 21, 11 3V3 in top + rail at number 21, 12 SDA in J19, 13 SCL in J20. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 14-17, AIR: 14 GND in top − rail at number 25, 15 3V3 in top + rail at number 25, 16 SDA in I19, 17 SCL in I20. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 18-21, YELLOW: 18 GND in bottom − rail at number 7, 19 HOLD in C5, 20 5V in bottom + rail at number 7, 21 LED− Y in D9. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 22-25, RED: 22 GND in bottom − rail at number 4, 23 DISC in A4, 24 5V in bottom + rail at number 4, 25 LED− R in A12. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 26-28, BUZZER: 26 GND in bottom − rail at number 12, 27 3V3 in C3, 28 BUZ in A8. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 29-32, NFC: 29 GND in top − rail at number 29, 30 3V3 in top + rail at number 29, 31 SDA in F19, 32 SCL in H20. Solder each wire from the top, then tie the bundle to the board edge.
- Leave the + rail link pads beside the left M3 hole open: the top + rail is 3V3 and the bottom + rail is 5V. Joining them would short 3V3 to 5V. Check with a meter: no short from 5V, 3V3 or SDA/SCL to GND, no short from 3V3 to 5V. Then plug U1 in, USB-C toward the board edge as drawn, after checking it is a power-left module. Plug U2 in with its screw terminal toward the top edge.

Parts fitted only when needed (marked DNP on the drawing): leave their holes free.

- Part 5, RSD1 (10 kΩ), top + rail at number 14 and I14: fit it only if the amplifier's SD pin reads near 0 V in the SD pull-up check.

No.	Item	Holes
	− rail link	right link pad, upper to right link pad, lower
	wire 3V3	E3 to top + rail at number 18 (along the channel)
	wire SCL	F2 to F20 (along the channel)
	wire LAMP R	F3 to E17 (along the channel)
	wire SDA	H9 to H19
	wire SDA	G1 to G9
	wire BCLK	J6 to J11
	wire DIN	I8 to I12
	wire GND	J15 to top − rail at number 15
	wire LRC	F7 to F10
	wire 5V	A1 to bottom + rail at number 1
	wire 5V	A16 to bottom + rail at number 16
	wire 5V	E16 to F16
	wire GND	A2 to bottom − rail at number 2
	wire GND	A11 to bottom − rail at number 11
	wire GND	A14 to bottom − rail at number 14
1	RG1 220 Ω	C7 to C10
2	RG2 220 Ω	C17 to C13
3	RP1 100 kΩ	B10 to bottom − rail at number 10
4	RP2 100 kΩ	B13 to bottom − rail at number 13
5	RSD1 10 kΩ (fit only if needed)	top + rail at number 14 and I14
6	Q1 BS170	D E9, G E10, S E11
7	Q2 BS170	D E12, G E13, S E14
8	U1 sockets	D1-D8, H1-H8

No.	Item	Holes
9	U2 socket	G10-G16
10-13	OLED wires	10 GND top − rail at number 21, 11 3V3 top + rail at number 21, 12 SDA J19, 13 SCL J20
14-17	AIR wires	14 GND top − rail at number 25, 15 3V3 top + rail at number 25, 16 SDA I19, 17 SCL I20
18-21	YELLOW wires	18 GND bottom − rail at number 7, 19 HOLD C5, 20 5V bottom + rail at number 7, 21 LED− Y D9
22-25	RED wires	22 GND bottom − rail at number 4, 23 DISC A4, 24 5V bottom + rail at number 4, 25 LED− R A12
26-28	BUZZER wires	26 GND bottom − rail at number 12, 27 3V3 C3, 28 BUZ A8
29-32	NFC wires	29 GND top − rail at number 29, 30 3V3 top + rail at number 29, 31 SDA F19, 32 SCL H20